



Pacing, graded **A**ctivity and **C**ognitive behaviour therapy: a randomised **E**valuation

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And many others

Background

Why was a trial needed?

- Systematic reviews concluded that rehabilitative cognitive behaviour and graded exercise therapies were the most promising treatments for CFS in secondary care but also that more research was needed.
- Large surveys by patient charities concluded that CBT and GET often made patients worse rather than better. Pacing and specialist medical care were reported to be more helpful.

The trial questions

Which of CBT, GET, pacing and SMC alone, are most effective?

Are any of the treatments more harmful than the others?

Methods

The Treatments

SMC

SMC + APT

SMC + CBT

SMC + GET

Specialist Medical Care

- Diagnosis
- Advice and education
 - Sleep, activity, rest
- Medication
- Self-help

Therapies

- Adaptive Pacing Therapy (APT)
- Cognitive Behaviour Therapy (CBT)
- Graded Exercise Therapy (GET)

Differences between therapies

- One adaptive (APT)
- Two - behavioural activation / graded exposure (CBT & GET)
- One addresses thoughts and feelings (CBT)

Design

- At least 3 sessions of SMC over 52 weeks
- 14 sessions of therapy over 23 weeks (+ booster session at 36 weeks)
- Outcome assessed at 12, 24 & 52 weeks

Primary outcomes

Fatigue – Chalder Fatigue Scale

Disability - SF36 physical functioning subscale

Results

Recruitment of sample

- 3,158 clinic attenders clinically assessed
- 898 research screened
- 641 recruited
- Approximately 160 per treatment group

Sample demographics

- Caucasian 93%
- Age (mean, SD) 38 (12)
- Female 78 %

Sample clinical characteristics

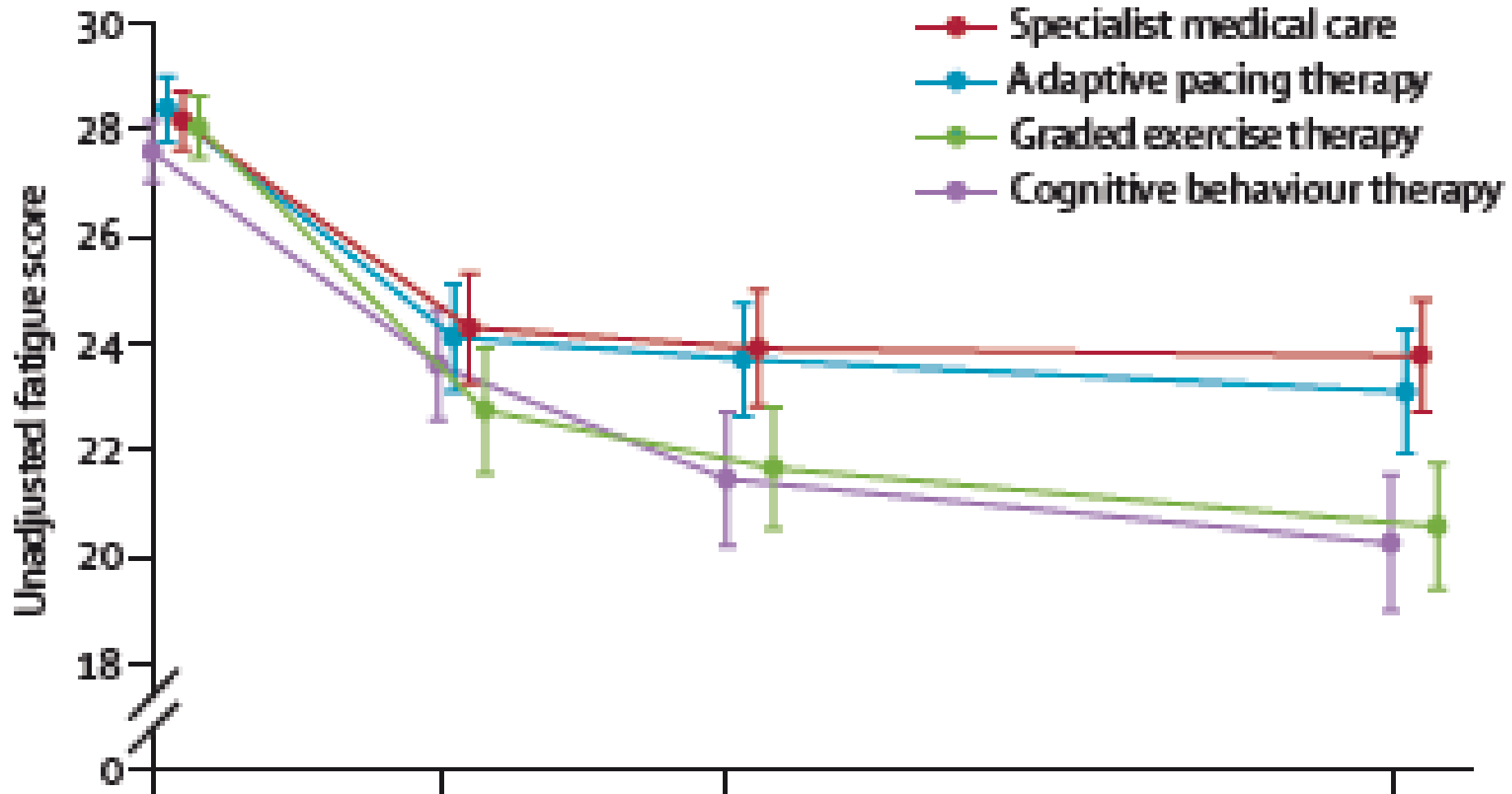
- CDC criteria 67%
- ME criteria 51%
- Current depressive disorder 33%
- Median duration of illness 32 months

- Mean (SD) CFS score 28 (3.8)
- Mean (SD) SF36 PF score 38 (16)

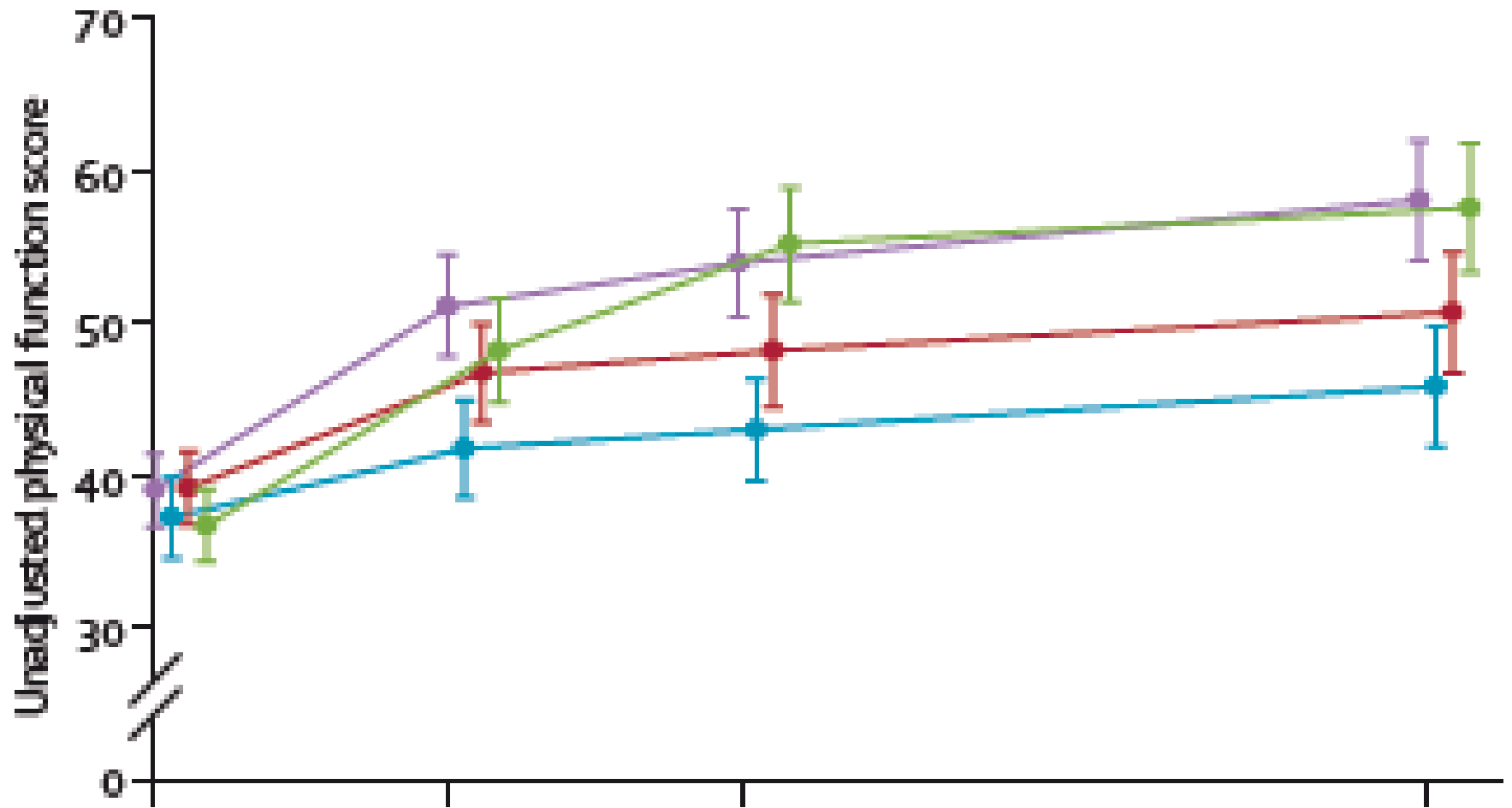
Drop-outs: N (%)

	SMC	APT	CBT	GET	All
No Rx received	1 (1)	1 (1)	3 (2)	0	5 (1)
Withdrawn from Rx	14 (9)	11 (7)	17 (11)	10 (6)	52 (8)
Lost to F.U.	8 (5)	6 (4)	13 (8)	6 (4)	33 (5)

A Fatigue: all participants



E Physical function: all participants



Primary outcomes models

- Baseline measures of outcome
- Time, time by intervention
- Stratification factors:
 - Centre
 - CDC criteria
 - London ME criteria
 - Depression diagnosis
- Cluster effects of therapist variation

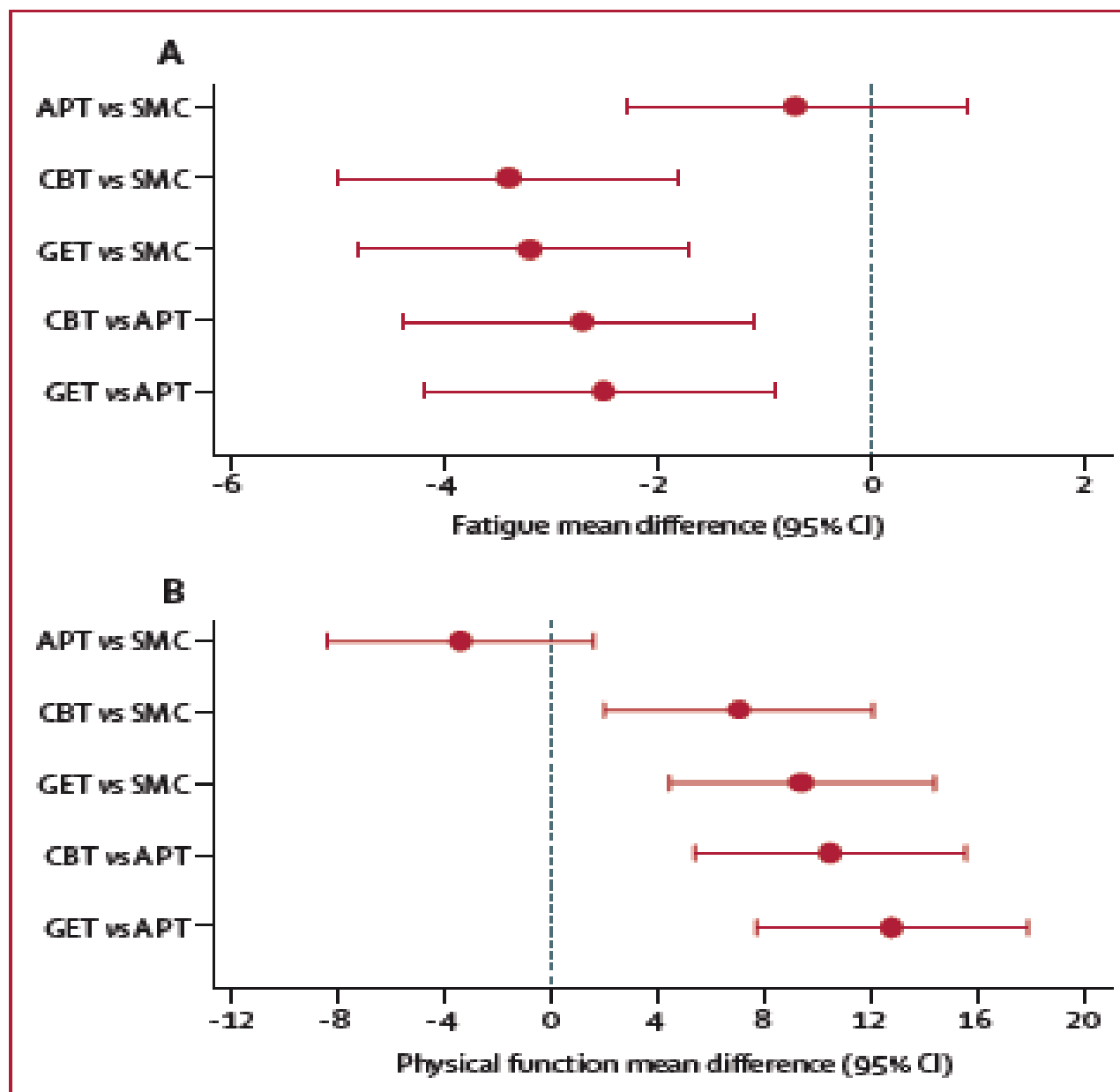


Figure 3: Primary outcome treatment differences for fatigue (A) and physical function (B) at 52 weeks

Clinically significant difference

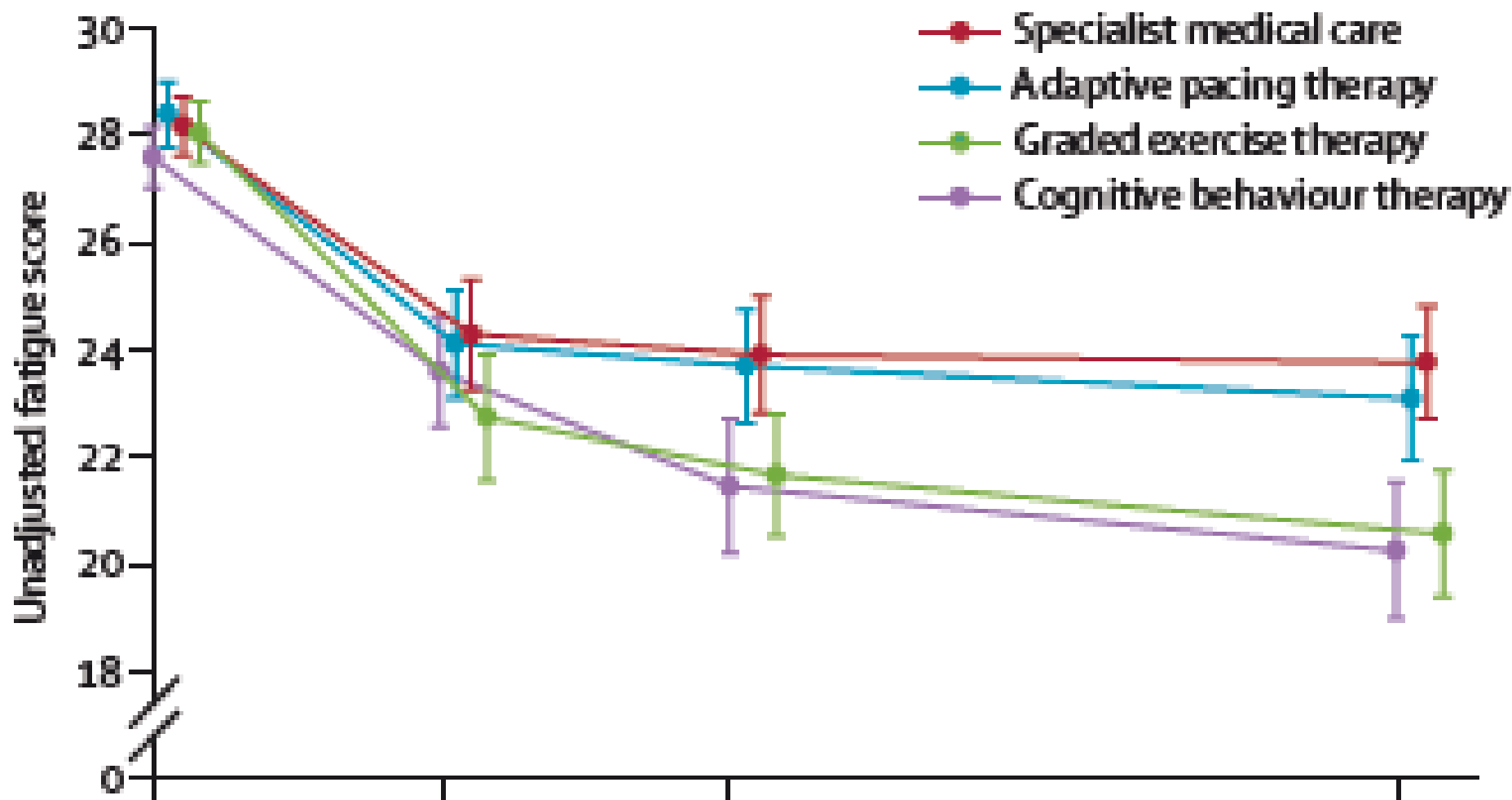
- Clinically significant difference (CUD) 0.5 SD of baseline scores
- Fatigue - 2 points
- Physical Function - 8 points

How effective?

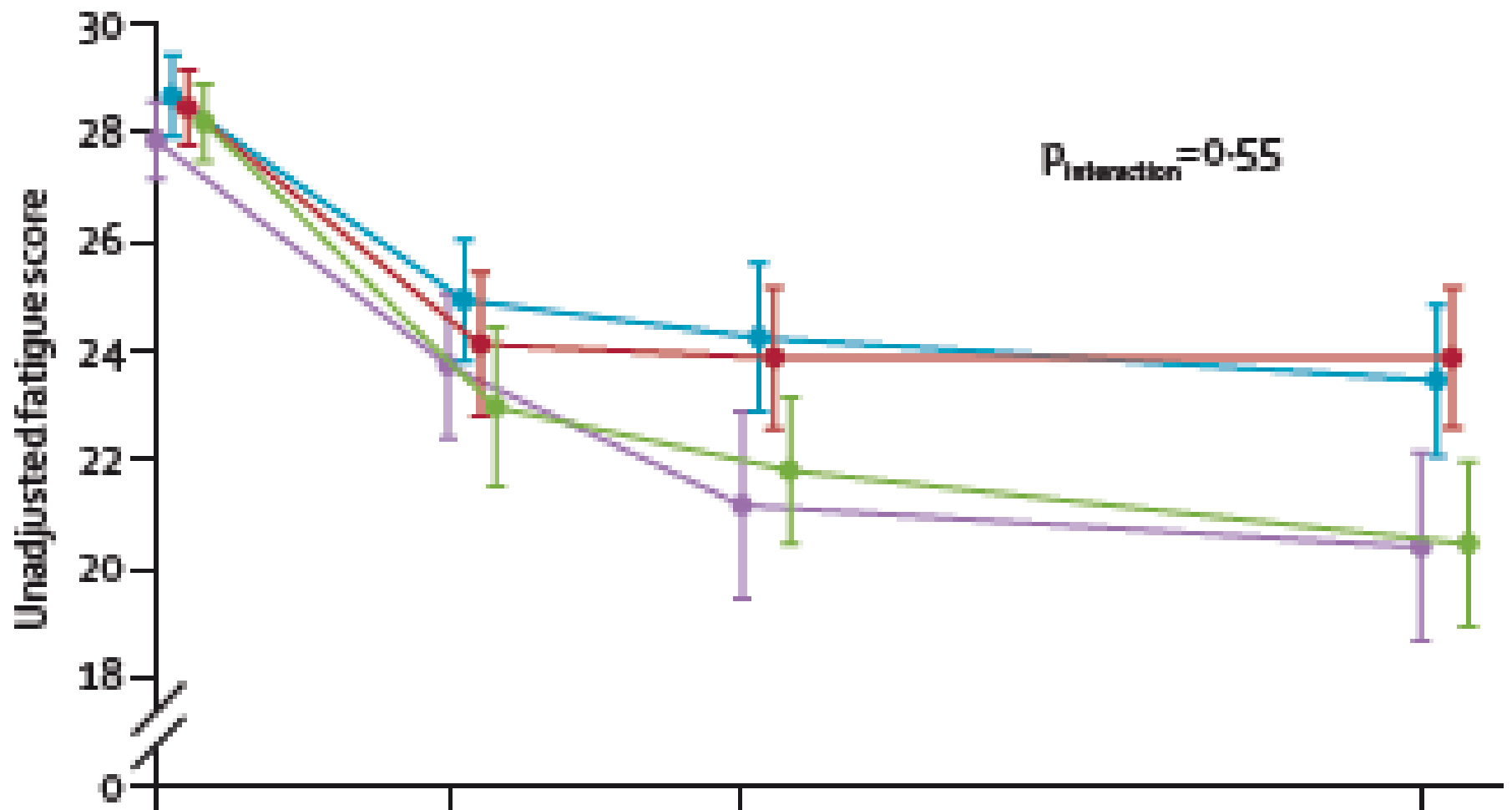
Percentage improved in both fatigue and function

	SMC	APT	CBT	GET
% improved	45	42	59	61
% “normal” levels	15	16	30	28

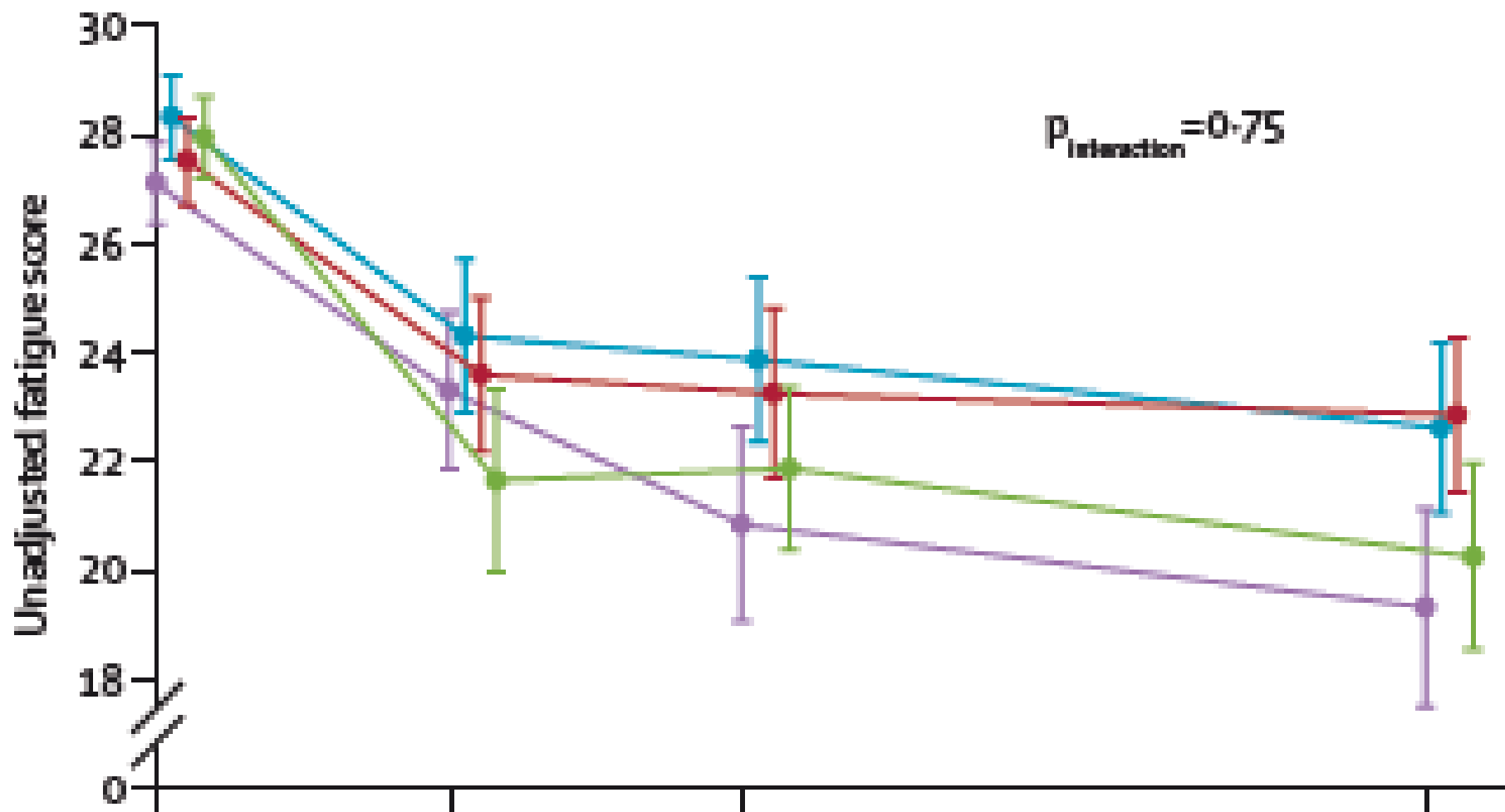
A Fatigue: all participants



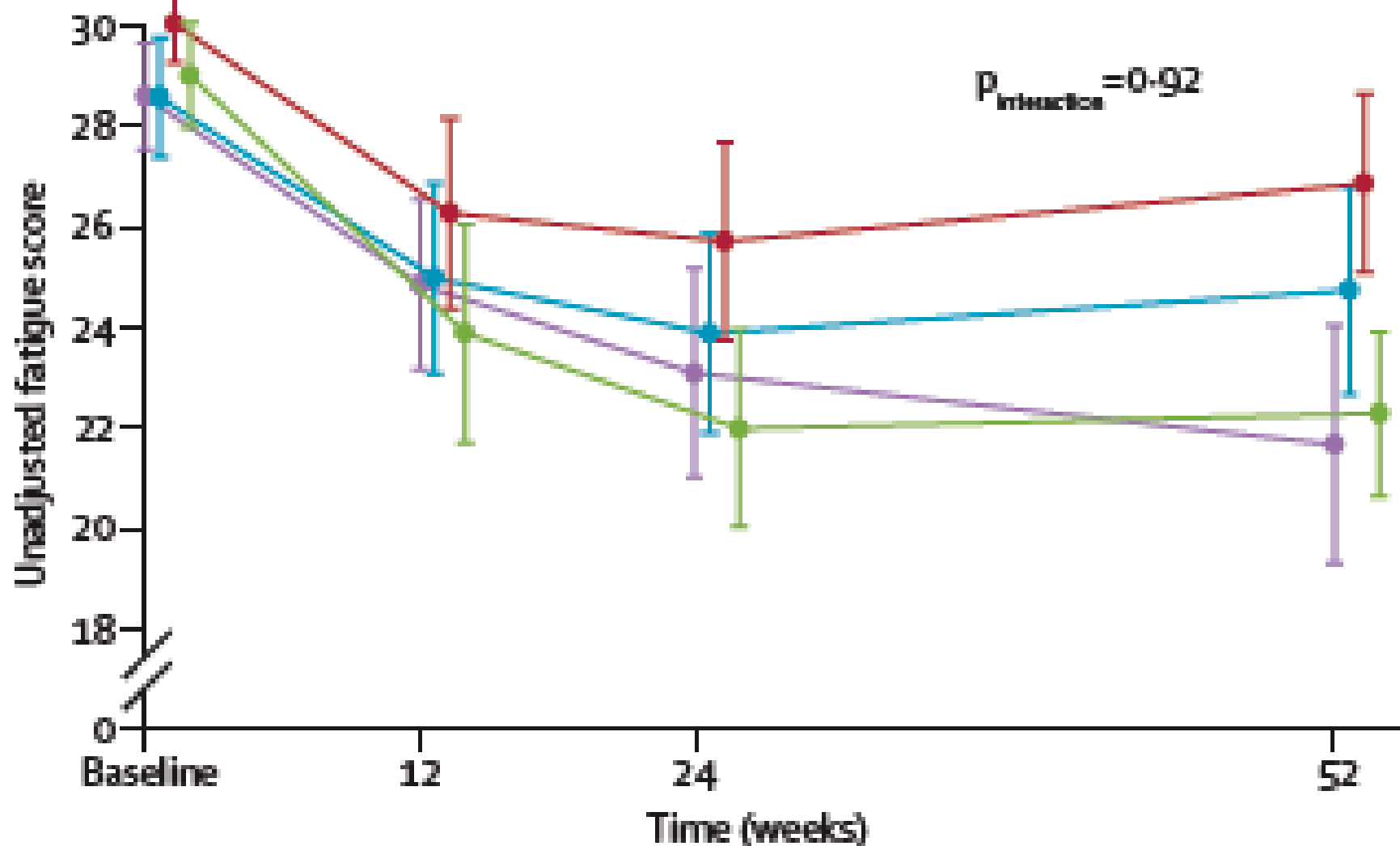
B Fatigue: international CFS only



C Fatigue: London ME only



D Fatigue: depressive disorder only



Clinical Global Impression

Overall, how much do you feel your health has changed since the start of the study? Please tick the one box below that most closely corresponds to how you feel now.

Very much better

Much better

A little better

No change

A little worse

Much Worse

Very much worse

CGI (2 levels) %

	SMC	APT	CBT	GET
“Much”+ better	25	31	41	41
“Much”+ worse	9	7	6	7

Other secondary outcomes

CBT and GET best:

Overall disability

Sleep disturbance

Post-exertional malaise

GET best:

Walking ability

CBT best:

Depression

Mixed picture or no better:

Anxiety, symptom count, poor concentration

Therapy quality

	APT	CBT	GET
N sessions	13	14	13
“Confident” before	72 %	57 %	70 %
“Satisfied” after	85 %	82 %	88 %
Alliance	6.5	6.5	6.5
Adherence	6	6	6.5

Safety

Measured 5 ways

% with events

	SMC	APT	GET	CBT
Serious adverse reactions	1	1	1	2
Serious deterioration	9	8	6	9

Conclusions

- CBT and GET are more effective than SMC alone and APT.
- APT is no different from SMC alone
- The effectiveness of CBT and GET is moderate
- The effect is similar however CFS/ME is defined and in those also depressed
- Treatments are safe, if given as described
- Pushing limits > staying within limits



Funders

